1. Pick two points distinct in ***R***, yet equivalent in the finite projective field $\mathbf{Z_3}$

Consider (1,1,1) and (2,2,2).
Cross product: [1 * 2 - 2*1 mod 3, 1 * 2 - 2*1 mod 3, 1 * 2 - 2*1 mod 3] = [0 mod 3,0 mod 3,0 mod 3] = [0,0,0].

2. Pick two unique points in $\mathbf{Z_3}$

Consider (1,0,0) and (0,1,0).
(1,0,0) x (0,1,0) = [0 * 0 - 0 * 1 mod 3, 0*0 - 1 * 0 mod 3, 1 * 1 - 0 * 0 mod 3] = [0,0,1], which is not equal to [0,0,0].

(0 * 1 + 0 * 0 + 1 * 0) mod 3 = 0, so (1,0,0) is on the line [0,0,1]
(0 * 0 + 0 * 1 + 1 * 0) mod 3 = 0, so (0,1,0) is on the line [0,0,1]

Name: generate_all_points
Input: mod - a positive integer prime modulus
Output: returns points - a sequence of unique points in the projective geometric space $Z_{mod,}$ where a point is a tuple of 3 integers

1. **points** ← an empty sequence
2. for each **first_value** in 0,1, …, **mod** - 1 do
    a. for each **second_value** in 0,1, … **mod** - 1 do
        i. **current_point** ← (1, **first_value**, **second_value**)
        ii. append **curent_point** at the end of **points**
3. for each **third_value** in 0,1, … **mod** - 1 do
    a. **current_point** ← (0, 1, **third_value**)
        i. append **curent_point** at the end of **points**
4. return **points**

Name: create_cards
Input:
- points - a sequence of unique points in the projective geometric space $Z_{mod,}$ where a point is a tuple of 3 integers
- lines- a sequence of unique lines in the projective geometric space $Z_{mod,}$ where a line is a tuple of 3 integers
- mod - a positive integer prime modulus

Output: card_list - a sequence of sequences of integers, where each nested sequence represents a unique spot it card

1. **cards** ← empty sequence
2. For each **line** in **lines** do
    a. **card** ← empty sequence
    b. for each **index** in 0,1, … , the length of **points** - 1 do

            i. if *incident*($\mathbf{points}_{\mathbf{index}}$, **line**, **mod**) then
                1. append $\mathbf{points}_{\mathbf{index}}$ at the end of **card**
        c. append **card** at the end of **cards**
3. return **cards**

Discussion

This is valid because you can make a valid spot it deck with as many cards as possible. To create a deck with 40 cards, you can use modulus 7 to create a deck with 57 valid spot it cards, the maximum possible with this modulus. Then, you can remove 17 cards to form a deck with 40 cards modulus 7 where all the properties hold.